

Maven Project To Connect To Web Servers Using Selenium Dependency

Maven Project for Practice Test Automation Website Using Selenium

A Maven project for a practice test automation website uses Selenium along with Apache Maven to automate testing of web applications.

In this project, Selenium is used to interact with the practice testing website by automatically opening the browser, navigating through web pages, entering data, clicking buttons, and verifying outputs. Maven manages all required Selenium dependencies and build processes through the pom.xml file.

The main purpose of this project is to perform automated testing on websites instead of manual testing. It helps testers check whether login pages, forms, buttons, and other website functionalities are working correctly.

Working Process

1. Maven downloads Selenium libraries automatically.
2. Selenium WebDriver launches the browser.
3. The browser connects to the practice automation website.
4. Test cases are executed automatically.
5. Results are verified and displayed.

Applications

- Login page testing
- Form validation testing
- Button and link testing
- Automated regression testing
- Web application quality assurance

Advantages

- Saves testing time
- Reduces human errors
- Improves testing efficiency
- Supports Continuous Integration using tools like Jenkins
- Easy project management using Maven

This project demonstrates how Maven and Selenium are used together for automated testing of practice web applications in DevOps and software testing environments.

Steps to execute Projects:

Step1:

Create a maven project on the terminal with the command:

```
$mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -  
DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

```

vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp01
vaishnavi@vaishnavi-VirtualBox: ~/devops21$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom -----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> mvn-archetype-plugin:1.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< mvn-archetype-plugin:1.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- mvn-archetype-plugin:1.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 1.6 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] -----
[INFO] Parameter: basedir, Value: /home/vaishnavi/devops21
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vaishnavi/devops21/MyMavenSeleniumApp01
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 12.417 s
[INFO] Finished at: 2020-03-04T21:36:23+05:30
[INFO] -----
vaishnavi@vaishnavi-VirtualBox: ~/devops21$ cd MyMavenSeleniumApp01
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp01$ ls
pom.xml  src
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp01$ google-chrome --version
Google Chrome 145.0.7632.110

```

The command can be broken down as:

DgroupId=com.example – project’s base package (reverse-domain style).

-DartifactId=MyMavenSeleniumApp – defines the name of the project (will also be the folder name).

-DarchetypeArtifactId=maven-archetype-quickstart – sets the Maven template to be the basic Java application.

-DinteractiveMode=false – deactivates interactive mode and uses the given values automatically.

Step 2:

Use \$cd command to go into the directory.

\$gedit pom.xml

pom.xml file is created to configure the Maven project. It defines the project information, adds required dependencies like Guava and Commons IO libraries, and includes build plugins to compile the Java code, run unit tests, and package the application into an executable JAR file.

```

<dependencies>
  <dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>3.8.1</version>
    <scope>test</scope>
  </dependency>

  <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->

```

```
<dependency>
  <groupId>org.seleniumhq.selenium</groupId>
  <artifactId>selenium-java</artifactId>
  <version>4.29.0</version>
</dependency>
</dependencies>

<!-- Build: Configuring plugins and build settings -->
<build>
  <plugins>
    <!-- Example: Maven Compiler Plugin to compile Java code -->
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.8.1</version>
      <configuration>
        <source>1.7</source>
        <target>1.7</target>
      </configuration>
    </plugin>
    <!-- Example: Maven Surefire Plugin to run tests -->
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-surefire-plugin</artifactId>
      <version>2.22.2</version>
    </plugin>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>

      <artifactId>maven-jar-plugin</artifactId>
      <version>3.0.1</version>
      <configuration>
        <archive>
          <manifestEntries>
            <Main-Class>com.example.App</Main-Class>
```

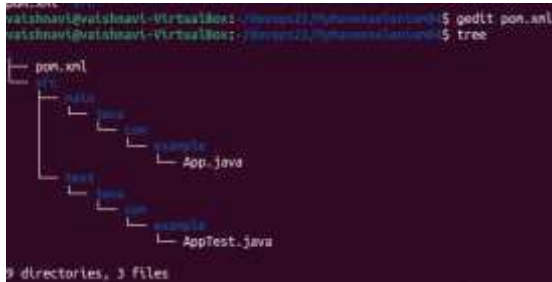
```

        </manifestEntries>
    </archive>
</configuration>
</plugin>

</plugins>
</build>
</project>

```

Execute \$tree command to view the structure of the project



```

neithnavi@neithnavi-VirtualBox: ~/openqa1/MyAutomationAutomation $ gedit pom.xml
neithnavi@neithnavi-VirtualBox: ~/openqa1/MyAutomationAutomation $ tree
.
├── pom.xml
├── src
│   ├── main
│   │   ├── java
│   │   │   └── com
│   │   │       └── example
│   │           ├── App.java
│   │           └── AppTest.java
│   └── test
│       ├── java
│       │   └── com
│       │       └── example
│       │           └── AppTest.java
└── 0 directories, 3 files

```

Step 3:

inspect the PracticeTestAutomation webpage to find the element IDs required for automation.

Right-click on the webpage and select Inspect. The HTML code will be displayed. Click on the cursor icon (arrow symbol) available on the top-left of the inspect panel, then hover over the username field to identify its ID, which is user-name.

Similarly, check the password field (ID: password) and the login button (ID:submit). Make a note of these IDs, as they will be used in the App.java file.

Next,

Edit the App.java file package

```
com.example;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.chrome.ChromeDriver;
```

```
/** * Hello world! */ public
```

```
class App
```

```
{
```

```
    public static void main( String[] args )
```

```
{
```

```
        WebDriver driver=new ChromeDriver();
```



```

vishal@kali:~/vishalnavi-virtualbox- /jwvcp2/PracticeTest2
[187] Listing for projects...
[188]
[189]
[190] Building dependency tree...
[191] Building dependency tree...
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```

Test login | Practice Test Automation

practice-test-automation.com

Chrome is being controlled by automated test software

Practice Test Automation

HOME PRACTICE COURSES BLOG CONTACT

Test login

This is a simple login page. Students can use this page to practice writing about positive and negative login methods, login functionality, something that most of the test automation engineers need to automate.

Default credentials to automate login:
Username: **student**
Password: **Password123**

Username

Password

Submit

A screenshot of a web browser displaying the 'Practice Test Automation' website. The browser's address bar shows the URL 'practicetestautomation.com/logged-in-successfully/'. Below the address bar, a message reads 'Chrome is being controlled by automated test software'. The website's header features the 'Practice Test Automation' logo on the left and a navigation menu with links for 'HOME', 'PRACTICE', 'COURSES', 'BLOG', and 'CONTACT' on the right. The main content area displays a large heading 'Logged In Successfully' followed by the text 'Congratulations student. You successfully logged in!'. At the bottom of this section is a dark button labeled 'Log out'.

Practice Test Automation Website

<https://practicetestautomation.com/practice-test-login/> is a sample website used for testing and learning web automation tools like Selenium.

It is used for:

- Writing automation scripts and UI tests
- Practicing login functionality testing
- Learning and improving test automation skills

Pushing The Project into Git:

Execute the commands:

```
$ git config --global user.name "vaishnavim1121"
```

```
$ git config --global user.email "vaishnavim1121@gmail.com"
```

```
$ git init
```

```
$ git add .
```

```
$ git commit -m "initial commit"
```

```
$ git remote add origin https://github.com/vaishnavim1121/MyMavenSeleniumApp.git
```

```
$ git remote set-url origin https://*PAT*@github.com/vaishnavim1121/MyMavenSeleniumApp
```

```
$ git push -u origin master
```

```

$ git init
$ git config --global user.name "vaishnavim1121"
$ git config --global user.email "vaishnavim1121@gmail.com"
$ git init
$ git add .
$ git commit -m "First commit"
[master (root-commit) 1a4a4b] First commit
13 files changed, 285 insertions(+)
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/example/App.java
 create mode 100644 src/test/java/com/example/AppTest.java
 create mode 100644 target/maven-status/maven-compiler-plugin/1.0-2007.jar
 create mode 100644 target/classes/com/example/App.class
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/TEST-com.example.AppTest.txt
 create mode 100644 target/surefire-reports/com.example.AppTest.txt
 create mode 100644 target/test-classes/com/example/AppTest.class
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/MyMavenSeleniumApp $ git config --global user.name "vaishnavim1121"
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/MyMavenSeleniumApp $ git config --global user.email "vaishnavim1121@gmail.com"
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/MyMavenSeleniumApp $ git remote add origin https://github.com/vaishnavim1121/MyMavenSeleniumApp.git
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/MyMavenSeleniumApp $ set-url origin https://ghs_wm55sgt2q0f0v7n0cjtja62V3w7f1qgithub.com/vaishnavim1121/MyMavenSeleniumApp
set-url: command not found
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/MyMavenSeleniumApp $ git remote set-url origin https://ghs_wm55sgt2q0f0v7n0cjtja62V3w7f1qgithub.com/vaishnavim1121/MyMavenSeleniumApp
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/MyMavenSeleniumApp $ git push -u origin master
remote: Repository not found.
fatal: repository 'https://github.com/vaishnavim1121/MyMavenSeleniumApp/' not found
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/MyMavenSeleniumApp $ git remote set-url origin https://ghs_wm55sgt2q0f0v7n0cjtja62V3w7f1qgithub.com/vaishnavim1121/MyMavenSeleniumApp
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/MyMavenSeleniumApp $ git push -u origin master
Enumerating objects: 39, done.
Counting objects: 100% (39/39), done.
Delta compression using up to 10 threads
Compressing objects: 100% (19/19), done.
Writing objects: 100% (39/39), 8.46 KiB | 721.00 KiB/s, done.
Total 39 (delta 0), reused 0 (delta 0), pack-reused 0
to https://github.com/vaishnavim1121/MyMavenSeleniumApp
 * [new branch] master -> master
Branch 'master' set up to track remote branch 'master' from 'origin'.
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/MyMavenSeleniumApp $

```

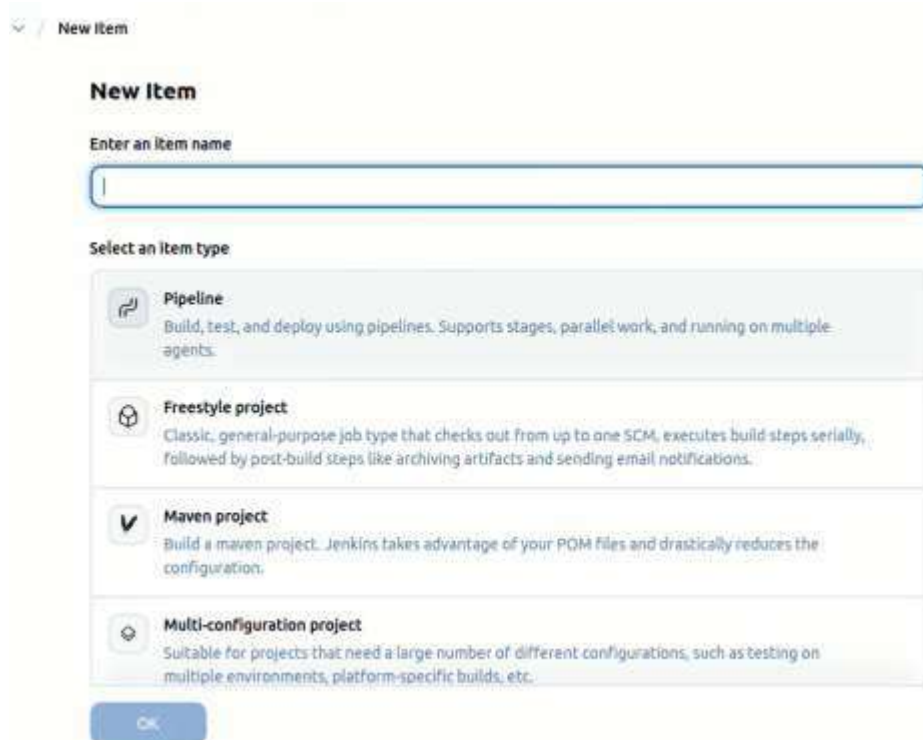
Second Phase

Depicting Jenkins Pipeline

Follow the given steps to deploy the project on Jenkins.

create a Jenkins pipeline job that will use the Jenkinsfile from your project repository

go to the Jenkins home page and click on “New Item” to create a job, enter the item name select Pipeline as the project type.



The screenshot shows the 'New Item' dialog box in Jenkins. At the top, there is a breadcrumb 'New Item'. Below it, the title 'New Item' is displayed. A text input field labeled 'Enter an Item name' is present. Underneath, a section titled 'Select an item type' lists four options: 'Pipeline' (with a description: 'Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.'), 'Freestyle project' (with a description: 'Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.'), 'Maven project' (with a description: 'Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.'), and 'Multi-configuration project' (with a description: 'Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.'). An 'OK' button is located at the bottom left of the dialog.

This option allows you to build, test, and deploy applications using a Jenkinsfile. After selecting the pipeline type, click “OK” to proceed to the job configuration page

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

https://github.com/vaishnavim1121/automation2.git

Please enter Git repository.

Credentials

github credentials

Advanced

Save Apply

Now we need to add a Jenkinsfile into our project which will consist of the operations that will occur when building the project in Jenkins

It includes step wise instructions guiding at each stage with right commands

```

1 pipeline {
2     agent any // Use any available agent
3
4     tools {
5         maven 'Maven' // Ensure this matches the name configured in Jenkins
6     }
7     stages {
8         stage('Checkout') {
9             steps {
10                 git branch: 'master', url: 'https://github.com/vaishnavim1121/SeleniumTestAutomation.git'
11             }
12         }
13
14         stage('Build') {
15             steps {
16                 sh 'mvn clean package' // Run Maven build
17             }
18         }
19
20         stage('Test') {
21             steps {
22                 sh 'mvn test' // Run unit tests
23             }
24         }
25
26         stage('Run Application') {
27             steps {
28
29             }
30         }
31     }
32 }

```

Now try building the project with Jenkins

Activities

Google Chrome

May 9 19:23

vaishnavim1121/Gradle

MyMavenSelenium02

localhost:8080/job/MyMavenSelenium02/

Jenkins

MyMavenSelenium02

Status

Changes

Build Now

Delete Pipeline

Configure

Rename

Full Stage View

Stages

Pipeline Syntax

MyMavenSelenium02

Add description

Stage View

Average stage times:
(full run time: ~55s)

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Test	Run Application	Declarative: Post Actions
Apr 13 18:35	1s	392ms	1s	27s	5s	11s	147ms

Builds

Filter

13 April 2025

#1 18:35

Permalinks

Last build (#1), 26 days ago

Last stable build (#1), 26 days ago

Last successful build (#1), 26 days ago

Last completed build (#1), 26 days ago

REST API

Jenkins 2.562

The project is built successfully.